AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-11-08 | Entry window 8 November 2026 (full day, Sunday) | 5 records screened in scope

5 RECORDS IN SCOPE (11 EXCLUDED)

3 OF 10 SUBTOPIC BANDS POPULATED

3 EFFECT ESTIMATES WITH A CI

0 TIER-A RECORDS

1 RECORDS CARRIED ON METADATA ALONE

Signal of the day

Indoor and outdoor PM act jointly on stroke in older Chinese adults: in CHARLS (13,917 adults, 1,060 strokes) high county-level $PM_{2.5}$ carried HR 1.25 (1.04–1.51), household solid fuel HR 1.38 (1.18–1.62), with a multiplicative interaction (p=0.022).

Ambient exposure is one pre-baseline year of the CHAP satellite-fused product averaged to county; solid fuel is a single questionnaire item. The ordinal joint-exposure score gives HR 1.10 (1.05–1.15) per step. **Where it is weak:** the $PM_{2.5}$ contrast is categorical with no cut-point or per-$\mu g/m^3$ estimate in the abstract; every resident of a county shares one value; no household PM was measured, so the "joint" category combines an ecological and a proxy exposure. **Why it matters here:** the interaction is the kind of claim paired indoor–outdoor low-cost nodes could test directly at the household, which neither exposure layer in this cohort can.

What changed today

- **A Sunday floor.** Five records from sixteen screened, three bands occupied (exposure assessment 2, sensing 2, cardiovascular 1), no tier-A record. Eighteen Europe PMC hits for 8 November had already been assigned to 6–7 November in the cross-day candidate set.
- **Personal sensors inside an intervention design.** URBANOME ran wearable and ambient monitoring, agent-based exposure modelling, spirometry, HRV and omics before and after co-created mobility, greening and traffic-calming interventions in nine European urban living labs. The abstract gives direction only (annoyance, sleep-breathing complaints down; FVC gradient attenuated) and a metabolite panel at AUROC ≈1.0 that the authors themselves call overoptimistic. No control arm.
- **Two sensing items for quality control and toxicity.** A labelled smart-home indoor air-quality dataset with experimentally induced contextual anomalies (Norway), useful for fault detection on unattended nodes if a PM channel is present — the abstract does not list channels. A review of automated field-deployable cell-exposure platforms maps hardware for toxicity-weighted sensing, without performance data.
- **Exposure basis (x1).** Of 5 records, 1 rests on a low-cost sensor class (the smart-home dataset, sensing side) and 1 on personal/wearable monitoring (URBANOME). The single health-side record (stroke) is *modelled* (satellite-fused, county mean); no health estimate today is tied to a measured exposure.

Window continuity

The 5 November issue closed with `last_entry_date` at 5 November. This issue was built on the **10 November 22:16 UTC (16:16 CST) run**, the 05:00 UTC slot firing 17 hours late; no run executed on 7–9 November. Before 22:00 the newest buildable day is 9 November, so this run **backfills 6, 7, 8 and 9 November as four issues** and builds the **W45 weekly (1–7 November)**, owed since Saturday. Sunday 8 November opens W46. A cross-day candidate set was built before authoring and each record assigned to its earliest day. No BACKFILL records this issue.

Corpus analytics

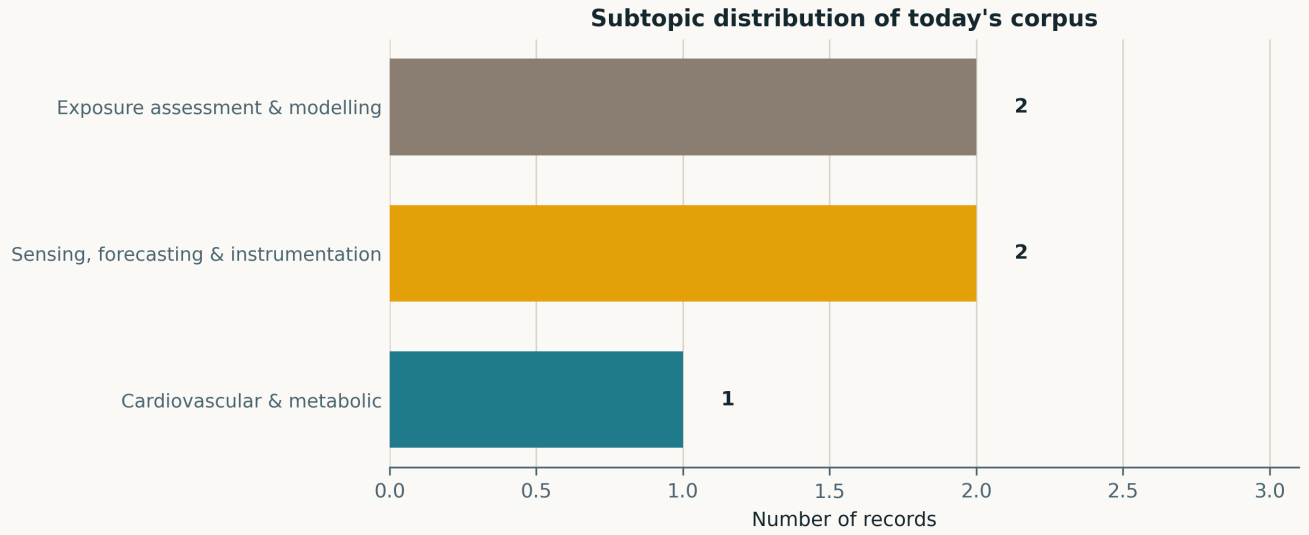


Fig. Subtopic assignment of the 5 records across three of ten bands: exposure assessment 2, sensing 2, cardiovascular and metabolic 1. Seven bands are empty, as expected on a Sunday entry day.

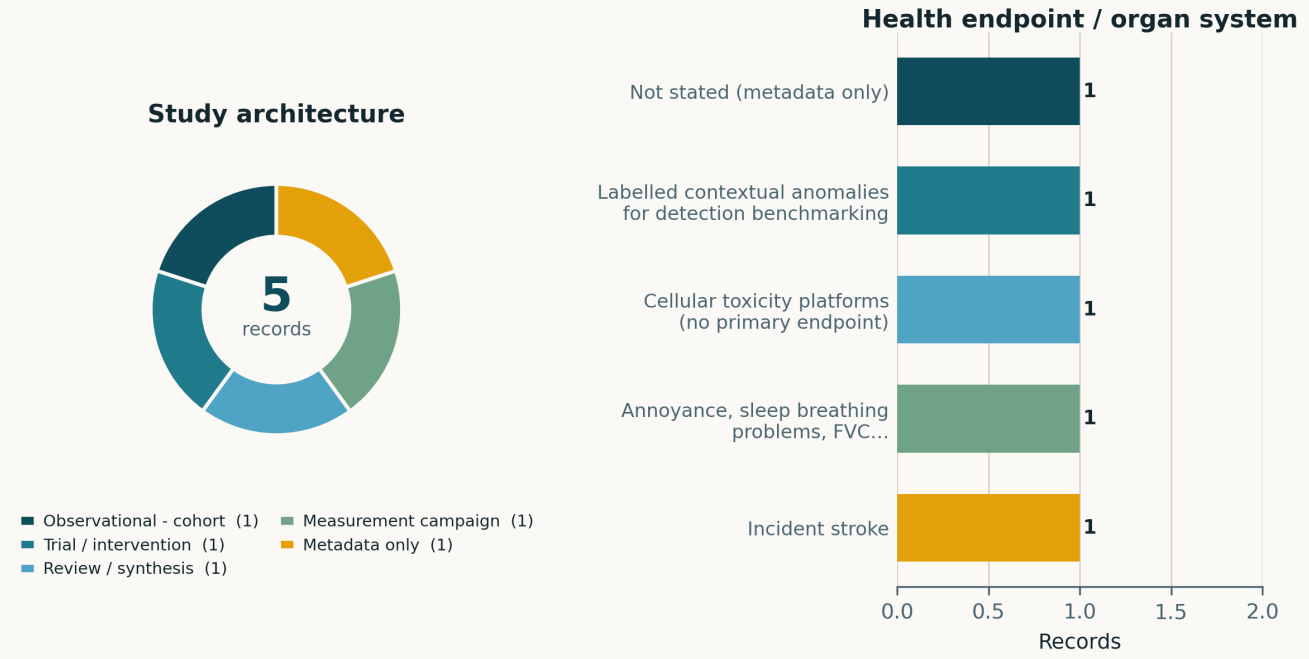


Fig. Left — architecture: one each of cohort, non-randomised intervention field study, review, data descriptor and metadata-only record. Right — every endpoint is a singleton; only incident stroke is a clinical outcome.

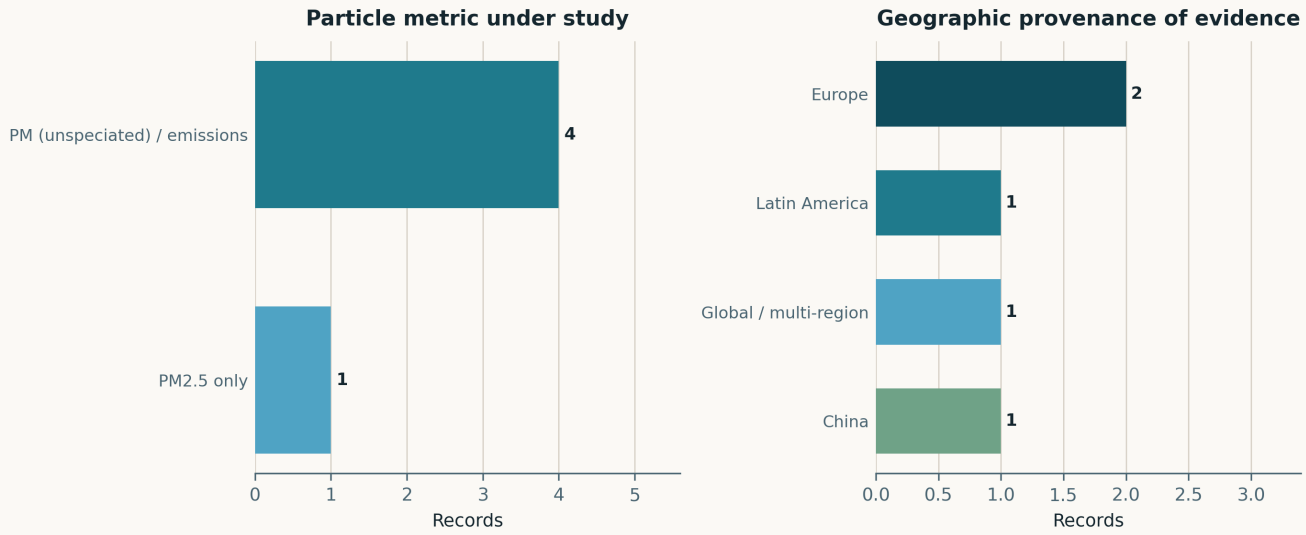


Fig. Left — particle metric as recorded per record; three records carry no single PM metric (review, data descriptor, metadata-only). Right — geography: China 1 (CHARLS), Brazil 1 (São Paulo State), one multi-region review and two European records (nine-city URBANOME; Norway).

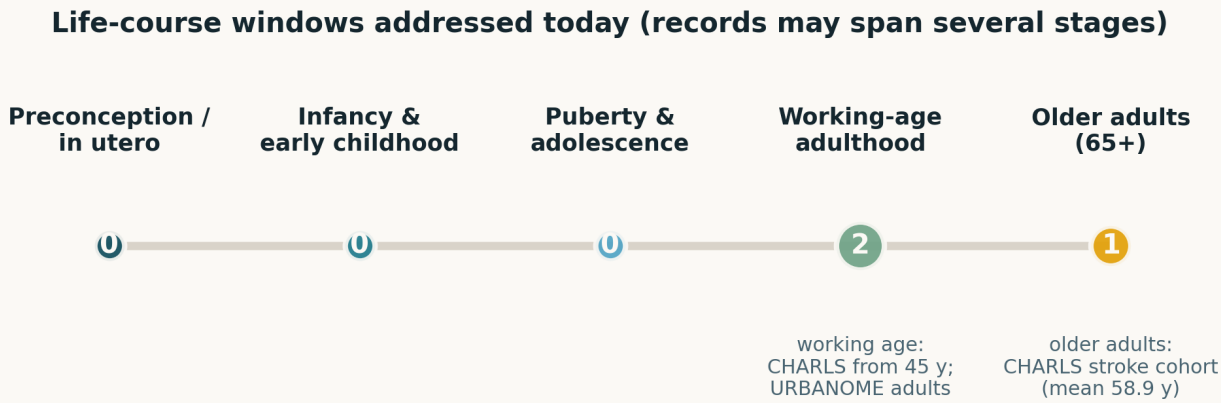


Fig. Life-course windows. Working age carries CHARLS (enrolment from 45 years) and URBANOME adults; older adults the CHARLS stroke cohort (mean age 58.9). No preconception, in-utero, childhood or adolescent record today.

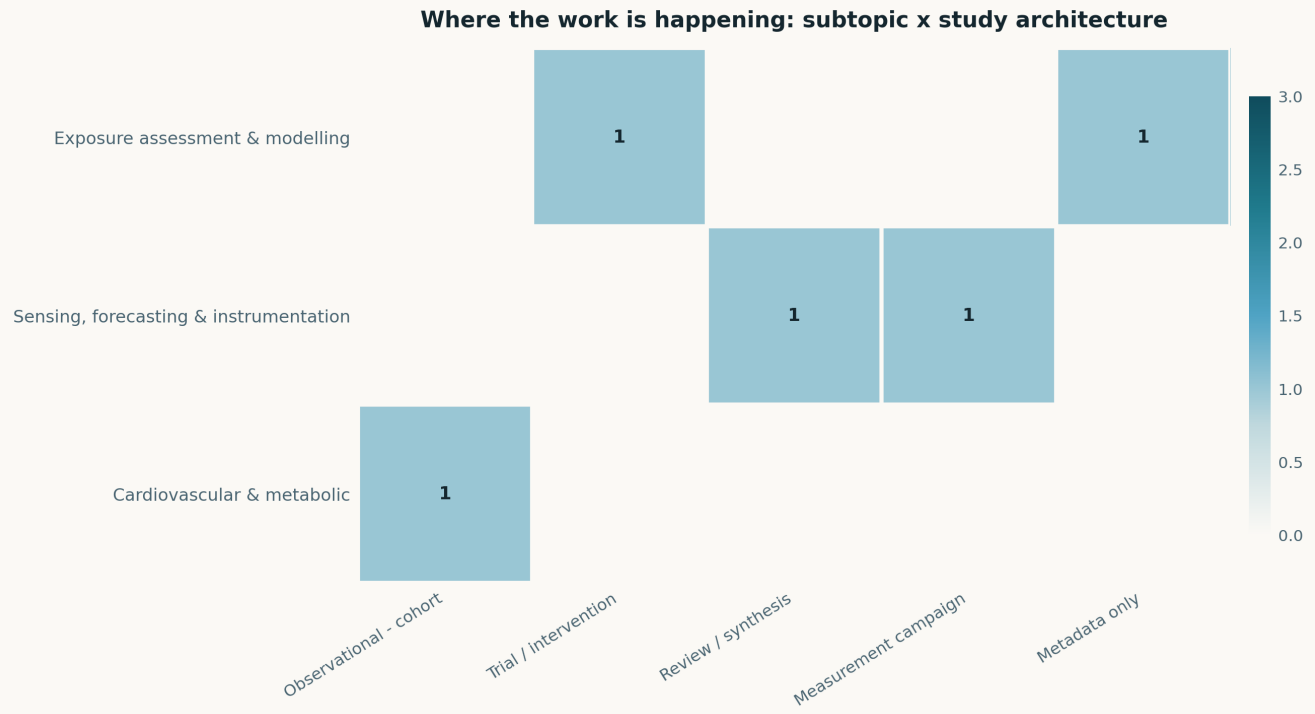


Fig. Subtopic × study architecture, five occupied cells, each with one record: the cohort sits in cardiovascular; exposure assessment holds the intervention study and the metadata-only record; sensing holds the review and the data descriptor.

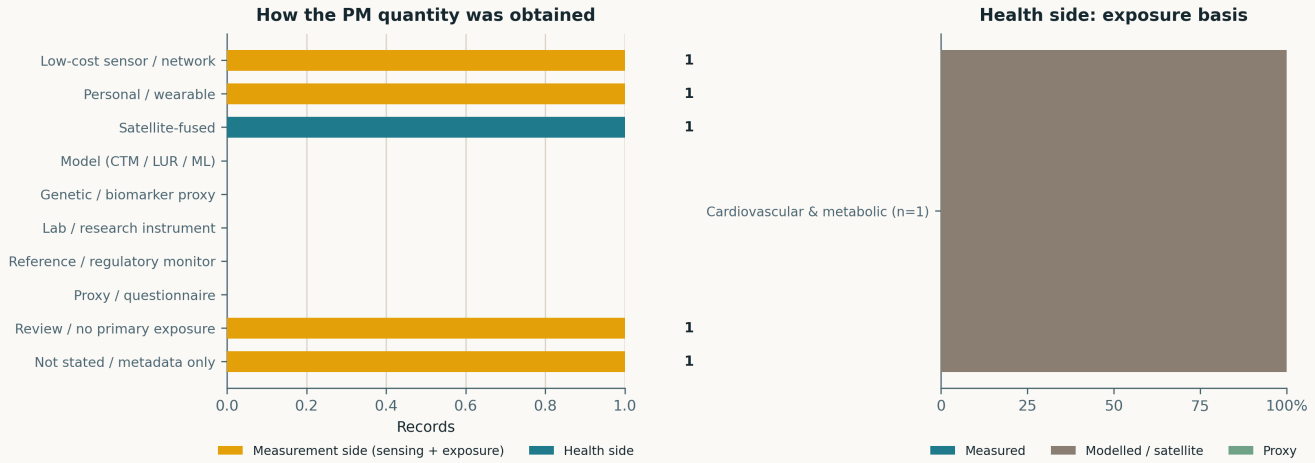


Fig. How each record obtained its PM quantity, classified from the exposure field. 1 of 5 records rest on a low-cost sensor or sensor network; 0 on reference or regulatory monitors. Health side, where an exposure basis is recoverable (n=1): measured 0 %, modelled or satellite-fused 100 %, proxy 0 %. Right panel sorted by modelled share: the subtopics at the bottom are where a measured low-cost network exposure would add most.

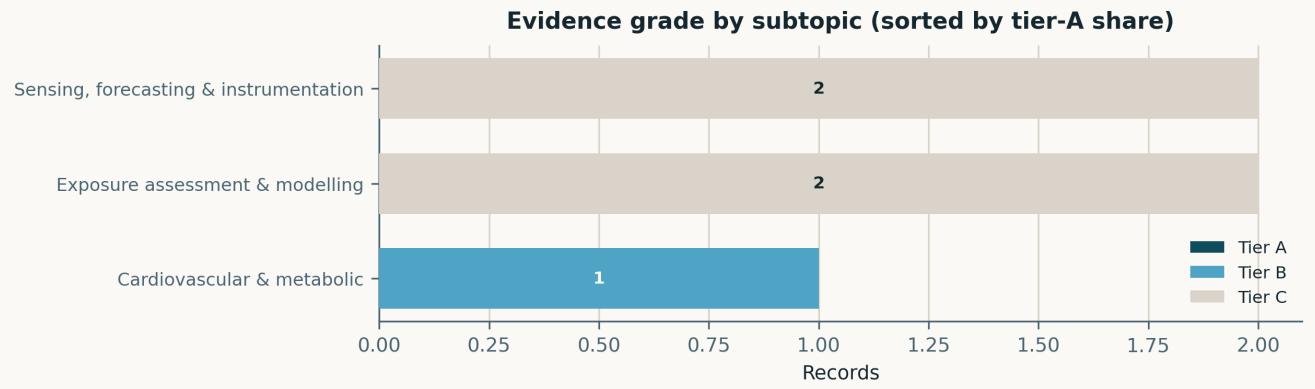


Fig. Tier mix per subtopic. 0 of 5 records are tier A (0 %). Highest tier-A share: *Sensing, forecasting & instrumentation* (0 of 2).

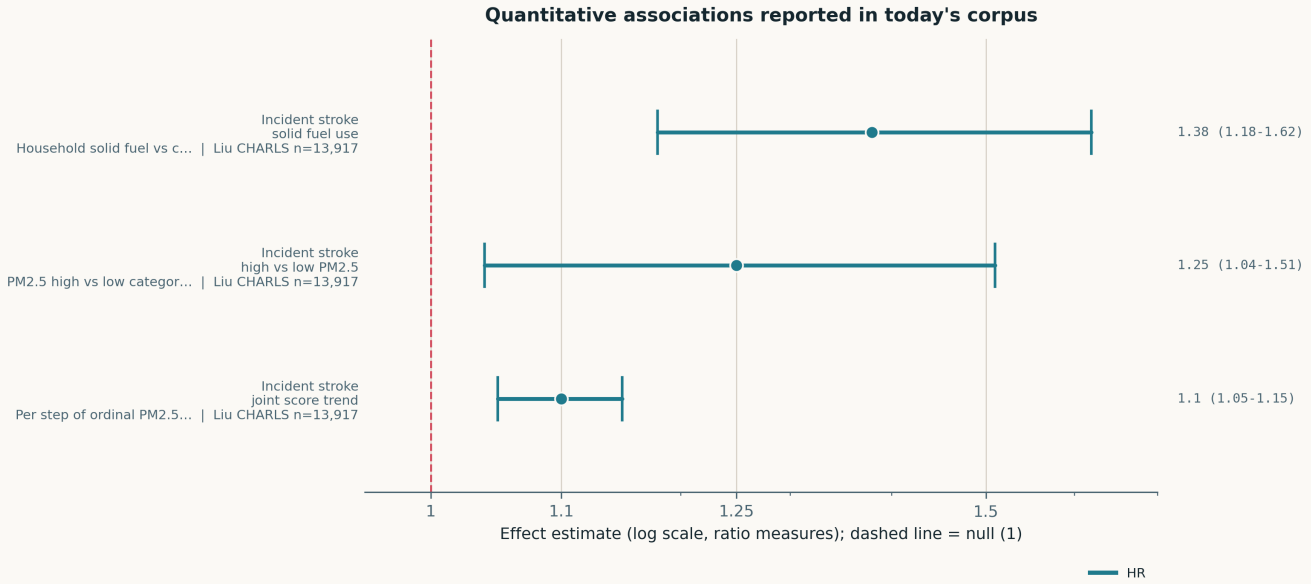


Fig. Ratio scale, 3 hazard ratios from one source (Liu, CHARLS, n=13,917): high vs low county $PM_{2.5}$, solid fuel vs clean fuel, and the per-step trend of the ordinal joint score. All three are categorical contrasts in the same sample and are not independent; no pooling and no I^2.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Cardiovascular & metabolic — 1 record

Liu CHARLS cohort county $PM_{2.5}$ solid fuel joint exposure incident stroke 2026 J Epidemiol Glob Health

Associations of PM(2.5) Exposure and Solid Fuel Use With Incident Stroke: Evidence From a Longitudinal Chinese Cohort.

CHARLS, 13,917 adults (mean age 58.9 y), 1,060 incident strokes. High vs low ambient $PM_{2.5}$: HR 1.25 (1.04-1.51); solid fuel users vs non-users: HR 1.38 (1.18-1.62); multiplicative interaction p=0.022; HR 1.10 (1.05-1.15) per step of an ordinal joint-exposure score. The $PM_{2.5}$ contrast is categorical and the cut-point and per-unit estimate are not in the abstract. County-level assignment from a satellite-fused product with one pre-baseline year gives every resident of a county the same value, and solid fuel is a one-time self-report with no measured household PM, so the joint category mixes an ecological and a proxy exposure. Relevance: indirect - indoor-outdoor co-exposure is exactly what paired household and ambient low-cost sensors could resolve.

doi: 10.1007/s44197-026-00623-x

Exposure assessment & modelling — 2 records

Sarigiannis URBANOME nine-city urban living labs exposome wearable sensors interventions 2026 Hum Genomics

URBANOME: an exposome-based framework linking urban interventions to molecular, health, and policy-relevant outcomes.

Horizon 2020 framework run in nine European urban living labs (Aarhus to Thessaloniki) evaluating co-created active-mobility, greening and traffic-calming interventions before and after, with wearable and ambient monitoring, agent-based exposure modelling, spirometry, blood pressure, HRV and omics. Pilot findings are qualitative: self-reported pollution annoyance and sleep-related breathing problems fell, pollution-related FVC differences attenuated in higher-exposure neighbourhoods; a four-metabolite panel separated pre/post samples with AUROC near 1.0, which the authors call likely overoptimistic. No PM concentration change, effect size or CI is reported in the abstract; designs, comparators and follow-up differ by site, and there is no control arm. Relevance: direct for method - an operational template for coupling personal sensors to intervention evaluation.

doi: 10.1186/s40246-026-01012-y

de Oliveira Sao Paulo State municipality air-quality anomalies COVID-19 restrictions ecological surveillance 2026 Atmos Pollut Res

Municipality-specific air-quality anomalies and exploratory ecological COVID-19 surveillance analyses in São Paulo State during 2020 restrictions

METADATA ONLY. Metadata only: Crossref deposit with no abstract on the day of entry and none recoverable by title lookup. From the title, the paper derives municipality-specific air-quality anomalies across Sao Paulo State during the 2020 COVID-19 restrictions and runs exploratory ecological analyses against COVID-19 surveillance data. No pollutant list, monitoring network, anomaly baseline, effect estimate or sample size can be stated, and no finding is asserted here. Any ecological pollution-COVID association from this period carries heavy confounding by mobility, testing intensity and season, which the full text will need to address. Relevance: indirect - lockdown anomaly baselines are a reference case for separating emission change from meteorology at municipal scale; recheck for the abstract in the weekly sweep.

doi: 10.1016/j.apr.2026.103258

Sensing, forecasting & instrumentation 2 records

Jeong review automated robotic field-deployable platforms aerosol cytotoxicity analysis 2026 iScience

Automated small-scale devices and robotic systems for analyzing the cellular impact of ambient aerosol particles.

Conceptual review of small, automated and robotic platforms that deliver ambient aerosol to cells for on-site cytotoxicity analysis, with emphasis on mobile and field-deployable systems, biomimetic device design, and the data acquisition and processing needed for a source-to-effect workflow. It reports no new measurements, no comparison of platforms against standard air-liquid-interface exposure, and no performance numbers in the abstract; search strategy and inclusion criteria are not described, so coverage cannot be judged. Relevance: indirect - toxicity-weighted sensing at the node is the logical step beyond mass concentration, and this is a map of the hardware candidates rather than evidence that any of them works in the field.

doi: 10.1016/j.isci.2026.118066

Toor SHoRT-CA smart home indoor air quality time series contextual anomalies dataset 2026 Data Brief

SHoRT-CA: Smart home data repository of time series with contextual anomalies.

Open repository of multivariate indoor air-quality time series from one IoT smart home, with human-activity and environmental-interaction records and contextual anomalies deliberately induced by experiments on the sensors, each documented and labelled for anomaly-detection benchmarking. The abstract does not list which pollutants or particle channels are measured, the sensor models, calibration, or record length, and a single dwelling with induced rather than naturally occurring faults limits how far detectors trained here transfer. Relevance: direct for quality control - labelled anomaly sets are scarce, and automated fault flagging is a standing need for unattended low-cost indoor networks, provided a PM channel is present.

doi: 10.1016/j.dib.2026.113370

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Liu CHARLS cohort county $PM_{2.5}$ solid fuel joint exposure incident stroke	Associations of PM(2.5) Exposure and Solid Fuel Use With Incident Stroke: Evidence From a Longitudinal Chinese Cohort.	*J Epidemiol Glob Health*	Prospective cohort	One-year pre-baseline county-level $PM_{2.5}$ from the China High Air Pollutants (CHAP) dataset; household solid cooking/heating fuel by questionnaire	China
Sensing, forecasting & instrumentation					
Jeong review automated robotic field-deployable platforms aerosol cytotoxicity analysis	Automated small-scale devices and robotic systems for analyzing the cellular impact of ambient aerosol particles.	*iScience*	Narrative review	Ambient aerosol particles sampled onto cell-based assays	Global / multi-region
Toor SHoRT-CA smart home indoor air quality time series contextual anomalies dataset	SHoRT-CA: Smart home data repository of time series with contextual anomalies.	*Data Brief*	Network deployment / data descriptor	Multivariate indoor air-quality time series from IoT smart-home sensors (channels not itemised in the abstract)	Norway
Exposure assessment & modelling					
Sarigiannis URBANOME nine-city urban living labs exposome wearable sensors interventions	URBANOME: an exposome-based framework linking urban interventions to molecular, health, and policy-relevant outcomes.	*Hum Genomics*	Intervention field trial (non-randomised)	Ambient and personal exposure monitoring (wearable sensors, smartphone apps) plus agent-based population exposure modelling	Europe
de Oliveira Sao Paulo State municipality air-quality anomalies COVID-19 restrictions ecological surveillance	Municipality-specific air-quality anomalies and exploratory ecological COVID-19 surveillance analyses in São Paulo State during 2020 restrictions	*Atmos Pollut Res*	Metadata only (no abstract)	Not stated (metadata only)	Brazil

Provenance and method

Entry window **8 November 2026**, single day (Sunday). Harvester legs: **PubMed** health **18**, sensing **2**; **Europe PMC 27**; **Crossref by ISSN 3**; **arXiv** no in-window entry; **OpenAlex** HTTP 429 on the harvester leg. PubMed connector (EDAT 6–9 November, health, instrumentation and broad MeSH axes) 69 PMIDs, **4 bucketed to 8 November, 2 appended**. The cross-day candidate set removed **18** records already assigned to 6–7 November (mostly Europe PMC re-surfacing), leaving **16 screened**. Abstracts: efetch filled 4 of 5 PMID records; Crossref title lookup recovered one EHP abstract; the APR deposit stayed empty. OpenAlex, alphaXiv, bioRxiv/medRxiv and Wiley Scholar Gateway connectors were not available on this session; **Scite** retraction check not run; no Amass backfill this run. Affiliations from PubMed XML. **5 admitted** (all window), **11 rejected** with logged reasons. **1 carried on metadata alone**. **Trial watch**: see the W45 weekly. Abstracts remain the copyright of their publishers.